

Supplementary materials for "ProteoformAnalyzer: a pipeline for assessing the impact of post-translational modifications and single-point mutations in proteins"

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Related Work

Computational analysis of proteoforms is primarily developed along different methodological lines, each addressing a specific part of the problem. A first category includes tools that assess the structural impact of missense variants. Pipelines such as **SNP2SIM** (3) reproducibly generate mutant structures and docking results to compare variant-induced alterations against the wild-type form, combining molecular dynamics and docking within a modular workflow. There is also **Mut-Map** (1), which maps mutations from the COSMIC database onto PDB and AlphaFold structures, superimposing them to quantify the structural context. However, such tools overlook post-translational modifications and are limited to monomeric representations.

It is also important to consider tools that model the effect of PTMs on structure and dynamics—such as **PTM-psi** (4), a Python package that applies post-translational modifications to protein structures and simulates their impact on dynamics and interactions, or **ProteoMeter** (6), which integrates data on multiple PTMs and limited proteolysis at single-residue resolution to understand the correlation between modifications and structure (though it operates on experimental proteomic data rather than generating or simulating structures de novo). Furthermore, language models capable of handling PTMs, such as **PTM-Mamba** (5), have recently extended ESM2 embeddings with modification-specific tokens to predict the functional effects of the PTMs themselves. However, these tools do not account for genetic variants, do not perform docking or ligand design, and remain limited to the single monomer, overlooking therapeutic aspects.

Previous work (2) introduced an integrated pipeline centered on transthyretin, combining structure prediction based on AlphaFold3, variant classification based on ESM2, docking with DiffDock-L and AutoDock Vina, and ligand optimization via DiffSBDD. Although effective, that framework was limited to TTR homotetramers, did not explicitly model PTMs, and lacked a user interface. As shown in Table 1, which compares tools operating on modeled or predicted structures or sequence, none of the existing approaches simultaneously combine variant analysis, PTM simulation, support for heteromeric oligomeric assemblies, docking, molecular dynamics, generative ligand design, and a graphical user interface. Proteoform Analyzer bridges this gap by unifying these functionalities into a single configurable tool scalable from monomer to tetramer, and sharing a common engine between the CLI and GUI.

Table 1. Summary of the features available in structure- and sequence-based proteoform analysis frameworks (mass-spectrometry-based identification tools are discussed separately in the text and omitted here, as they operate on a different class of input data). Var, analysis of missense variants; PTM, structural simulation of post-translational modifications; Het, support for heterotetrameric assemblies; Dock, molecular docking; MD, molecular dynamics; Des, generative ligand and antibody design; GUI, graphical user interface.

Tools	Var	PTM	Het	Dock	MD	Des	GUI
SNP2SIM (3)	Yes	No	No	Yes	Yes	No	No
Mut-Map (1)	Yes	No	No	No	No	No	Yes
PTM-Psi (4)	Yes	Yes	No	Yes	Yes	No	No
PTM-Mamba (5)	No	Yes	No	No	No	No	No
ProteoMeter (6)	No	Yes	No	No	No	No	No
ProteoformAnalyzer	Yes	Yes	Yes	Yes	Yes	Yes	Yes

1. Complete benchmark results

This section reports the complete per-proteoform outputs of the three benchmark experiments described in the main text—the TTR missense-variant panel, the TTR PTM panel, and the hemoglobin variant panel—complementing the top-ten summary figures of the main manuscript with full-length tables and the complete set of per-module plots generated by the workflow. All values are reported as produced by the pipeline, without manual post-processing; failed stages are reported as failures rather than interpolated. Docking scores are AutoDock Vina estimates, $\Delta\Delta G$ values are ThermoMPNN predictions, and the Proteoform Impact Score (PIS) is the composite defined in the main text, with degenerate components dropped and weights renormalized as described there.

1.1. Missense variants of the TTR homotetramer

Table 2 reports the complete set of 87 missense variants of uncertain significance, sorted by decreasing Proteoform Impact Score. For each variant the table lists the TM-score against the wild-type reference structure, the ThermoMPNN $\Delta\Delta G$ prediction, the tafamidis AutoDock Vina affinity in the transferred thyroxine-binding pocket, the end-of-trajectory backbone RMSD of the short OpenMM validation trajectory, the number of Louvain communities detected in the protein contact network, and the four non-degenerate PIS components with the composite score. Across the panel, TM-scores span 0.946–0.990 (the wild-type reference scores 1.000 by definition), predicted $\Delta\Delta G$ spans -0.36 to $+3.29$ kcal/mol, docking affinities span -5.63 to -4.60 kcal/mol (wild type: -5.04 kcal/mol), and the composite PIS spans 0.087–0.601. Trajectories completed for 83 of the 87 variants; the four failures (D38G, E109Q, R124C, and Y136V) are marked as missing in the RMSD column and their dynamics component was set to the panel-neutral value by the workflow. The number of Louvain communities ranges from 8 to 13 across the analyzed structures (wild type: 10).

In addition to the 87 single mutants, the proteoform stage generated two combinatorial proteoforms pairing a variant with a phosphorylation site (E74G with pSER72 and E74K with pSER72). These were scored by the impact module (composite 0.059 and 0.000, respectively, with TM-score 1.000) but were not docked, and are therefore omitted from Table 2.

Figures 1–9 show the complete per-module plots for this experiment: the full Proteoform Impact ranking of all scored proteoforms, the complete tafamidis docking score distribution, the full TM-score summary, the RMSD and RMSF overlays of the molecular dynamics trajectories, the pocket volume and drift profiles, and the RFAntibody ranking and interface-quality plots.

Table 2: Complete results for the 87 missense variants of uncertain significance on the TTR homotetramer (UniProt P02766), sorted by decreasing Proteoform Impact Score. TM-score: structural similarity to the wild-type reference tetramer (TM-align). $\Delta\Delta G$: ThermoMPNN prediction (kcal/mol). Affinity: AutoDock Vina score for tafamidis in the transferred thyroxine-binding pocket (kcal/mol). RMSD: end-of-trajectory backbone RMSD of the short OpenMM trajectory (Å); missing entries denote trajectories that failed and were reported as such. k_{Lou} : number of Louvain communities in the protein contact network. The last four numeric columns are the PIS components and the composite score; the structural and sequence components were degenerate for this panel and the composite was computed from the four listed components with renormalized weights. The wild-type reference row is shown for comparison.

Variant	TM-score (–)	$\Delta\Delta G$ (kcal/mol)	Affinity (kcal/mol)	RMSD (Å)	k_{Lou} (–)	c_{struct} (–)	c_{bind} (–)	c_{dyn} (–)	c_{net} (–)	PIS (–)
A65D	0.957	2.00	-5.55	18.2	12	0.04	0.87	0.70	0.67	0.601
F53L	0.966	1.00	-5.63	17.3	11	0.03	1.00	0.66	0.33	0.576
F53I	0.960	1.14	-5.63	17.8	10	0.04	1.00	0.68	0.00	0.523
G67R	0.979	2.85	-4.60	17.9	11	0.02	0.74	0.69	0.33	0.487
E81G	0.956	1.14	-5.44	19.0	11	0.04	0.68	0.73	0.33	0.480
F53V	0.970	1.35	-5.56	17.8	10	0.03	0.88	0.68	0.00	0.479
A117S	0.959	2.29	-5.53	18.6	10	0.04	0.83	0.71	0.00	0.472
S43N	0.959	0.64	-4.82	23.1	12	0.04	0.37	0.89	0.67	0.467
S70R	0.979	0.65	-4.76	18.2	12	0.02	0.47	0.70	0.67	0.451
T80A	0.978	1.40	-5.41	18.0	11	0.02	0.63	0.69	0.33	0.449
G67A	0.958	2.27	-4.76	23.3	11	0.04	0.47	0.89	0.33	0.446
Y136S	0.978	2.24	-5.40	23.8	10	0.02	0.62	0.91	0.00	0.440
E74G	0.957	1.45	-5.28	18.6	12	0.04	0.41	0.71	0.67	0.438
L75P	0.974	2.58	-5.39	17.8	11	0.03	0.59	0.68	0.33	0.434
N144S	0.956	0.15	-5.45	18.9	10	0.04	0.70	0.72	0.00	0.427
T69A	0.986	2.27	-4.92	19.3	13	0.01	0.19	0.74	1.00	0.422
T126N	0.980	2.02	-5.03	26.1	13	0.02	0.01	1.00	1.00	0.418
S72P	0.972	2.08	-5.44	17.7	10	0.03	0.68	0.68	0.00	0.407
V91A	0.971	2.47	-4.85	18.8	12	0.03	0.31	0.72	0.67	0.404
E62G	0.954	1.45	-5.40	18.1	10	0.05	0.62	0.70	0.00	0.392

Table 2: (continued)

Variant	TM-score (-)	$\Delta\Delta G$ (kcal/mol)	Affinity (kcal/mol)	RMSD (Å)	k_{Lou} (-)	c_{struct} (-)	c_{bind} (-)	c_{dyn} (-)	c_{net} (-)	PIS (-)
Y89H	0.971	2.77	-5.20	17.9	12	0.03	0.28	0.69	0.67	0.387
A140S	0.961	0.46	-4.79	18.6	11	0.04	0.41	0.71	0.33	0.381
R54T	0.965	2.52	-5.31	23.1	10	0.03	0.46	0.89	0.00	0.380
I104S	0.972	1.50	-5.27	18.5	9	0.03	0.39	0.71	0.33	0.370
T139M	0.958	0.18	-5.16	18.8	12	0.04	0.20	0.72	0.67	0.370
G67E	0.983	2.93	-5.27	18.3	11	0.02	0.39	0.70	0.33	0.367
K55N	0.956	2.46	-4.84	19.8	11	0.04	0.33	0.76	0.33	0.365
A65T	0.952	-0.06	-5.15	18.5	12	0.05	0.19	0.71	0.67	0.364
I93V	0.962	0.95	-5.00	23.3	12	0.04	0.07	0.89	0.67	0.360
S97Y	0.987	0.62	-5.04	19.5	13	0.01	0.01	0.75	1.00	0.358
F53C	0.961	1.22	-5.23	18.2	11	0.04	0.34	0.70	0.33	0.350
V50M	0.976	1.62	-5.23	18.1	11	0.02	0.32	0.69	0.33	0.342
T79K	0.957	1.87	-4.93	22.7	11	0.04	0.18	0.87	0.33	0.337
L32P	0.979	2.75	-4.97	18.6	12	0.02	0.11	0.71	0.67	0.331
E81K	0.970	0.85	-5.11	18.0	12	0.03	0.12	0.69	0.67	0.330
M33I	0.973	-0.36	-4.98	18.9	12	0.03	0.09	0.73	0.67	0.326
A117G	0.960	1.71	-5.08	18.5	12	0.04	0.07	0.71	0.67	0.321
K90N	0.970	2.41	-5.28	18.1	10	0.03	0.42	0.69	0.00	0.318
I127V	0.960	1.00	-4.99	18.2	12	0.04	0.07	0.70	0.67	0.318
L78R	0.952	2.46	-5.02	18.6	12	0.05	0.03	0.71	0.67	0.306
G121S	0.971	1.23	-5.01	18.4	12	0.03	0.04	0.70	0.67	0.305
G73E	0.946	3.29	-5.03	18.2	8	0.05	0.02	0.70	0.67	0.300
L75Q	0.956	2.90	-5.03	18.3	12	0.04	0.02	0.70	0.67	0.300
V50A	0.970	2.10	-5.04	19.0	12	0.03	0.01	0.73	0.67	0.298
E109K	0.960	0.46	-5.06	23.8	11	0.04	0.04	0.91	0.33	0.298
F64S	0.957	2.62	-4.94	18.9	11	0.04	0.16	0.72	0.33	0.297
V48M	0.954	0.95	-4.95	18.9	11	0.05	0.15	0.72	0.33	0.293
D38E	0.956	2.47	-5.02	23.7	11	0.04	0.03	0.91	0.33	0.292
G26S	0.981	0.95	-5.13	18.9	11	0.02	0.17	0.72	0.33	0.292
V50L	0.990	1.06	-4.93	24.8	10	0.01	0.18	0.95	0.00	0.292
A65S	0.958	0.25	-5.13	18.3	11	0.04	0.15	0.70	0.33	0.288
D94H	0.956	1.37	-4.85	18.2	10	0.04	0.32	0.70	0.00	0.288
V40I	0.973	0.03	-4.96	19.0	11	0.03	0.13	0.73	0.33	0.283
W61L	0.956	2.38	-4.99	20.2	11	0.04	0.09	0.77	0.33	0.282
I127M	0.969	2.04	-4.96	19.1	11	0.03	0.12	0.73	0.33	0.281
H110N	0.971	0.54	-4.97	18.8	11	0.03	0.12	0.72	0.33	0.277
R124H	0.961	2.01	-5.10	19.0	11	0.04	0.11	0.73	0.33	0.277
V50G	0.955	2.76	-4.97	18.6	11	0.04	0.11	0.71	0.33	0.275
E62D	0.956	0.42	-5.09	19.0	11	0.04	0.10	0.73	0.33	0.275
P122R	0.970	0.30	-4.97	18.1	11	0.03	0.12	0.70	0.33	0.270
D58A	0.954	1.36	-4.98	18.6	11	0.05	0.09	0.71	0.33	0.270
V142I	0.973	-0.17	-4.97	18.2	11	0.03	0.12	0.70	0.33	0.270
Y134H	0.973	0.39	-4.98	19.0	11	0.03	0.09	0.73	0.33	0.268
V142A	0.972	1.58	-4.97	18.0	11	0.03	0.11	0.69	0.33	0.267
G67V	0.960	2.31	-4.99	18.8	11	0.04	0.08	0.72	0.33	0.265
A111S	0.984	0.65	-4.98	18.6	11	0.02	0.09	0.71	0.33	0.262
F84L	0.957	1.48	-5.00	18.1	11	0.04	0.07	0.70	0.33	0.257
V114A	0.988	1.66	-4.98	17.6	11	0.01	0.09	0.68	0.33	0.253
A56P	0.962	1.50	-5.00	17.8	11	0.04	0.07	0.68	0.33	0.252
D58V	0.973	2.00	-5.02	19.6	11	0.03	0.03	0.75	0.33	0.252
A129T	0.976	0.28	-5.01	18.5	11	0.02	0.05	0.71	0.33	0.249
Y134C	0.983	-0.05	-5.05	19.7	11	0.02	0.02	0.76	0.33	0.247
L131M	0.961	0.40	-5.04	18.6	11	0.04	0.01	0.71	0.33	0.241
T69I	0.972	1.75	-5.14	18.8	10	0.03	0.18	0.72	0.00	0.240
L78H	0.956	2.23	-5.04	18.3	11	0.04	0.01	0.70	0.33	0.238

Table 2: (continued)

Variant	TM-score (—)	$\Delta\Delta G$ (kcal/mol)	Affinity (kcal/mol)	RMSD (Å)	k_{Lou} (—)	c_{struct} (—)	c_{bind} (—)	c_{dyn} (—)	c_{net} (—)	PIS (—)
Y98F	0.974	1.08	-5.02	23.3	10	0.03	0.02	0.90	0.00	0.224
S70I	0.953	1.18	-4.96	18.0	10	0.05	0.12	0.69	0.00	0.216
I104T	0.983	1.19	-4.96	18.3	10	0.02	0.13	0.70	0.00	0.215
P44S	0.972	0.94	-5.07	18.9	10	0.03	0.06	0.73	0.00	0.198
C30R	0.969	1.10	-5.08	17.7	10	0.03	0.08	0.68	0.00	0.197
I88L	0.956	0.40	-5.08	17.4	10	0.04	0.08	0.67	0.00	0.196
I104N	0.960	1.55	-5.04	18.8	10	0.04	0.01	0.72	0.00	0.182
E74K	0.985	1.29	-5.02	18.1	10	0.02	0.03	0.69	0.00	0.179
Y136V	0.955	1.38	-4.89	—	11	0.05	0.25	—	0.33	0.157
D38G	0.955	2.80	-4.89	—	11	0.04	0.24	—	0.33	0.154
E109Q	0.954	0.30	-4.99	—	9	0.05	0.08	—	0.33	0.097
R124C	0.955	2.12	-5.01	—	11	0.04	0.05	—	0.33	0.087
<i>Wild-type reference:</i>										
WT	1.000	—	-5.04	18.2	10	—	—	—	—	—

1.2. PTM proteoforms of the TTR homotetramer

Table 3 reports the execution status of the fourteen analytical stages for the PTM experiment, complementing the stage table of the TTR variant experiment in the main text. The proteoform and $\Delta\Delta G$ stages were skipped because this experiment contains no missense variants, and the ESM2 stage completed the embedding step but the UMAP projection failed for this single-sequence run; both outcomes are reported explicitly by the workflow.

Table 4 reports the complete results for the wild-type tetramer and the seven single-site PTM proteoforms. Because each modified structure was generated from the same Boltz-2 wild-type prediction, all TM-scores are exactly 1.000, and the structural and sequence PIS components were degenerate and dropped with weight renormalization, so the composite reflects binding, dynamics, and network evidence only. Tafamidis affinities span the narrow range -5.05 to -5.01 kcal/mol (wild type: -5.02 kcal/mol), consistent with the modification sites lying outside the transferred thyroxine-binding pocket. Short OpenMM trajectories completed for the wild-type tetramer and the pTHR23 and pSER72 proteoforms (final RMSD 17.98–18.31 Å); the remaining five simulations terminated with non-finite particle positions and are marked as failed rather than interpolated, so their dynamics component is missing and the composite was computed from the available components.

Figures 12–18 show the complete per-module plots for this experiment.

Figure 11 presents the betweenness-centrality values and Louvain community assignments for each residue in the V50M-TTR tetramer, whereas Figure ?? illustrates the residue-level differences in centrality between the V50M and wild-type TTR tetramers.

Table 3. Execution status for the TTR PTM experiment (seven single-site modifications on the wild-type tetramer).

Status	Step	Elapsed (s)	N outputs	Message
OK	sequence	1.8	3	Retrieved 1 sequences, 0 mutations, 1 consequence tables; mature chain: P02766 21-147
OK	structure	1.8	1	Boltz API prepared 1 structures
OK	ptm	1.1	7	Generated 7 PTM structures (7 site(s) × 1 structures; 0 site/structure combos unverifiable)
SKIP	proteoform	0.0	0	proteoform_mode='single' — no combinatorial proteoforms
OK	tmscore	3.5	3	Computed TM-scores for 8 structures (engine=binary)
OK	pcn	27.3	2	PCN analysis on 8 structures (centrality + communities)
OK	md	271.7	19	MD (openmm): 3 ok, 0 skipped, 8 total
OK	esm2	10.4	1	ESM2 embeddings done (1 seqs) but UMAP failed: n_components=2 must be between 0 and min(n_samples, n_features)=1 with svd_solver='full'
OK	pocket	3.7	2	Pocket prediction: 8 structures, 8 with real predicted sites (methods: reference)
OK	docking	22.5	26	Vina docking (single): 8/8 docked, 1 ligand(s)
OK	ligand_design	8.2	7	DiffSBDD generated 1 ligand file(s) for wt-P02766-tetramer (pocket: ['A:14', 'A:107', 'A:71', 'A:13', 'A:15', 'A:108', 'A:30', 'A:73']); 5 target+ligand complex(es) written for the 3D viewer
SKIP	ddg	0.0	0	No mutations available
OK	impact_score	0.0	2	Computed impact scores for 7 proteoforms (top: wt-P02766-tetramer_phosphorylation_THR23 = 0.585); used components: binding, dynamics, network; dropped degenerate (no local signal): structural, sequence
OK	antibody	3008.7	8	RFAntibody complete (local quiver CLI): collected 40 designed complex(es) against 6 hotspot(s). Filter by RF2 pAE<10.0, RMSD<2.0Å (see scores.tsv).

Table 4. Complete results for the TTR PTM experiment. TM-score: structural similarity to the wild-type reference (TM-align). Affinity: AutoDock Vina score for tafamidis in the transferred thyroxine-binding pocket (kcal/mol). RMSD: end-of-trajectory backbone RMSD of the short OpenMM trajectory (Å); “failed (NaN)” denotes trajectories terminated with non-finite particle positions, reported as produced by the workflow. The structural and sequence PIS components were degenerate for this panel (all zeros) and dropped with weight renormalization; missing dynamics components correspond to failed trajectories. p, phosphorylation; n, nitrosylation.

Proteoform	TM-score (—)	Affinity (kcal/mol)	RMSD (Å)	c _{struct} (—)	c _{bind} (—)	c _{dyn} (—)	c _{net} (—)	PIS (—)
WT	1.000	-5.02	17.98	—	—	—	—	—
nCYS10	1.000	-5.04	failed (NaN)	0.00	0.17	—	0.67	0.232
pSER137	1.000	-5.04	failed (NaN)	0.00	0.21	—	1.00	0.328
pSER72	1.000	-5.03	18.31	0.00	0.09	1.00	0.00	0.349
pTHR138	1.000	-5.04	failed (NaN)	0.00	0.16	—	0.33	0.151
pTHR23	1.000	-5.05	17.99	0.00	0.28	0.98	0.67	0.585
pTHR25	1.000	-5.01	failed (NaN)	0.00	0.05	—	0.67	0.177
pTYR125	1.000	-5.04	failed (NaN)	0.00	0.15	—	0.00	0.069

1.3. Missense variants of the hemoglobin heterotetramer

Table 5 reports the execution status of the fourteen analytical stages for the hemoglobin experiment. All stages configured for this variant-only run completed successfully, including the molecular dynamics stage for all 51 structures.

Table 6 reports the complete set of 50 clinically relevant HBA and HBB substitutions, sorted by decreasing Proteoform Impact Score, with the same columns used for the TTR variant panel plus the affected subunit. TM-scores against the wild-type reference span 0.990–0.999 (wild type: 1.000), docking affinities for voxelotor in the pocket transferred from PDB 5E83 span -7.65 to -7.00 kcal/mol (wild type: -7.21 kcal/mol), the composite PIS spans 0.175–0.601, and the number of Louvain communities ranges from 10 to 14 (wild type: 13). ThermoMPNN $\Delta\Delta G$ predictions were returned for 27 of the 50 substitutions, each scored in the two equivalent subunit contexts of the assembly (54 predictions with a maximum within-variant discrepancy below 0.01 kcal/mol); the table reports the mean of the two contexts, and the remaining 23 substitutions, all on HBB, are marked as missing. Trajectories completed for all 50 variants.

Figures 19–27 show the complete per-module plots for this experiment.

Table 5. Execution status for the hemoglobin experiment (50 clinically relevant HBA/HBB substitutions on the $\alpha_2\beta_2$ heterotetramer, 51 Boltz-2 structures).

Status	Step	Elapsed (s)	N outputs	Message
OK	sequence	2.6	5	Retrieved 2 sequences, 50 mutations, 2 consequence tables; mature chain: P69905 2-142; P68871 2-147
OK	structure	45.3	51	Boltz API prepared 51 structures
SKIP	ptm	0.0	0	run_ptm=False
SKIP	proteoform	0.0	0	proteoform_mode='single' — no combinatorial proteoforms
OK	tmscore	111.2	3	Computed TM-scores for 51 structures (engine=binary)
OK	pcn	240.0	2	PCN analysis on 51 structures (centrality + communities)
OK	md	3471.5	307	MD (openmm): 51 ok, 0 skipped, 51 total
OK	esm2	65.1	4	Embedded 52 sequences with ESM2-650M + UMAP
OK	pocket	29.4	2	Pocket prediction: 51 structures, 51 with real predicted sites (methods: reference)
OK	docking	388.7	155	Vina docking (single): 51/51 docked, 1 ligand(s)
OK	ligand_design	12.2	7	DiffSBDD generated 1 ligand file(s) for Mut.P68871.A11D-tetramer (pocket: ['A:102', 'A:62', 'A:66', 'A:105', 'A:101', 'A:129', 'A:63', 'A:25']); 5 target+ligand complex(es) written for the 3D viewer
OK	ddg	13.7	1	$\Delta\Delta G$ predictions: 54/54 mutations scored via thermomppn
OK	impact_score	0.1	2	Computed impact scores for 50 proteoforms (top: Mut.P68871.W16G-tetramer = 0.601); used components: structural, binding, dynamics, network; dropped degenerate (no local signal): sequence
OK	antibody	3502.5	8	RFAntibody complete (local quiver CLI): collected 40 designed complex(es) against 6 hotspot(s). Filter by RF2 pAE<10.0, RMSD<2.0A (see scores.tsv).

Table 6: Complete results for the 50 clinically relevant missense variants on the hemoglobin $\alpha_2\beta_2$ heterotetramer (HBA, UniProt P69905; HBB, UniProt P68871), sorted by decreasing Proteoform Impact Score. Columns as in Table 2, with the affected subunit listed next to the variant. $\Delta\Delta G$ values are the mean of the two equivalent subunit contexts where available (27 substitutions); missing entries denote substitutions for which no ThermoMPNN prediction was returned. The wild-type reference row is shown for comparison.

Variant	Subunit	TM-score (—)	$\Delta\Delta G$ (kcal/mol)	Affinity (kcal/mol)	RMSD (Å)	k_{Lou} (—)	c_{struct} (—)	c_{bind} (—)	c_{dyn} (—)	c_{net} (—)	PIS (—)
W16G	HBB (β)	0.997	—	-7.60	17.0	11	0.00	0.89	0.72	0.67	0.601
S10C	HBB (β)	0.999	—	-7.49	17.1	10	0.00	0.62	0.72	1.00	0.565
A6D	HBA (α)	0.991	-0.04	-7.59	17.6	12	0.01	0.87	0.74	0.33	0.541
K17N	HBA (α)	0.999	1.08	-7.58	17.6	12	0.00	0.83	0.74	0.33	0.527
D7Y	HBA (α)	0.998	1.70	-7.51	23.1	12	0.00	0.67	0.98	0.33	0.526
D7A	HBA (α)	0.997	1.48	-7.65	17.1	13	0.00	1.00	0.72	0.00	0.524
N10T	HBA (α)	0.997	0.30	-7.48	17.2	11	0.00	0.60	0.73	0.67	0.501
A20D	HBA (α)	0.991	0.11	-7.54	17.3	12	0.01	0.75	0.73	0.33	0.499
E7K	HBB (β)	0.994	—	-7.53	17.2	12	0.01	0.72	0.73	0.33	0.487
V2A	HBB (β)	0.998	0.27	-7.53	17.4	12	0.00	0.72	0.74	0.33	0.486
K9T	HBB (β)	0.999	—	-7.60	17.3	13	0.00	0.88	0.73	0.00	0.482
D7V	HBA (α)	0.994	1.73	-7.51	18.0	12	0.01	0.67	0.76	0.33	0.477
H3Q	HBB (β)	0.992	—	-7.42	17.1	11	0.01	0.47	0.72	0.67	0.457

Table 6: (continued)

Variant	Subunit	TM-score (—)	$\Delta\Delta G$ (kcal/mol)	Affinity (kcal/mol)	RMSD (Å)	k_{Lou} (—)	c_{struct} (—)	c_{bind} (—)	c_{dyn} (—)	c_{net} (—)	PIS (—)
K8E	HBA (α)	0.995	1.24	-7.48	17.6	12	0.01	0.61	0.75	0.33	0.450
L15P	HBB (β)	0.999	—	-7.46	23.6	13	0.00	0.57	1.00	0.00	0.437
L3R	HBA (α)	0.993	2.34	-7.46	17.6	12	0.01	0.56	0.75	0.33	0.434
Y25H	HBA (α)	0.993	1.72	-7.53	17.4	13	0.01	0.72	0.74	0.00	0.430
K9Q	HBB (β)	0.992	—	-7.45	16.7	14	0.01	0.54	0.71	0.33	0.417
A13D	HBA (α)	0.995	0.53	-7.51	17.1	13	0.00	0.67	0.72	0.00	0.406
H3Y	HBB (β)	0.998	—	-7.42	17.6	12	0.00	0.47	0.75	0.33	0.402
K9E	HBB (β)	0.999	—	-7.49	17.2	13	0.00	0.62	0.73	0.00	0.391
P6R	HBB (β)	0.994	—	-7.40	18.3	14	0.01	0.41	0.77	0.33	0.389
H21Q	HBA (α)	0.994	0.09	-7.46	17.3	13	0.01	0.56	0.73	0.00	0.373
H21R	HBA (α)	0.991	-0.46	-7.39	17.4	14	0.01	0.39	0.74	0.33	0.373
V12D	HBB (β)	0.998	—	-7.30	17.2	11	0.00	0.19	0.73	0.67	0.357
E8G	HBB (β)	0.999	—	-7.07	17.0	12	0.00	0.34	0.72	0.33	0.347
A14P	HBA (α)	0.990	2.01	-7.00	17.5	13	0.01	0.48	0.74	0.00	0.345
G19R	HBA (α)	0.996	0.96	-7.34	17.0	14	0.00	0.28	0.72	0.33	0.330
D7N	HBA (α)	0.993	1.20	-7.03	17.5	13	0.01	0.43	0.74	0.00	0.328
E7Q	HBB (β)	0.999	—	-7.32	17.8	14	0.00	0.23	0.75	0.33	0.319
A14D	HBB (β)	0.998	1.75	-7.39	17.7	13	0.00	0.40	0.75	0.00	0.318
G19D	HBA (α)	0.998	0.90	-7.32	17.0	12	0.00	0.23	0.72	0.33	0.310
A20E	HBA (α)	0.999	0.16	-7.36	17.9	13	0.00	0.34	0.76	0.00	0.298
G16R	HBA (α)	0.993	-0.11	-7.15	18.5	12	0.01	0.14	0.78	0.33	0.293
L15R	HBB (β)	0.998	—	-7.15	23.5	13	0.00	0.14	1.00	0.00	0.285
H3R	HBB (β)	0.999	—	-7.16	17.8	14	0.00	0.12	0.76	0.33	0.280
K17M	HBA (α)	0.993	0.21	-7.34	17.5	13	0.01	0.28	0.74	0.00	0.274
E8K	HBB (β)	0.998	—	-7.26	17.6	12	0.00	0.10	0.74	0.33	0.270
V2E	HBA (α)	0.995	0.69	-7.26	16.7	12	0.00	0.12	0.71	0.33	0.267
A11D	HBB (β)	0.994	—	-7.24	17.9	12	0.01	0.05	0.76	0.33	0.255
A11V	HBB (β)	0.992	—	-7.24	17.0	12	0.01	0.07	0.72	0.33	0.253
W16R	HBB (β)	0.999	—	-7.20	18.1	12	0.00	0.03	0.77	0.33	0.251
K12E	HBA (α)	0.999	0.82	-7.13	17.9	13	0.00	0.18	0.76	0.00	0.242
E7A	HBB (β)	0.999	—	-7.28	16.9	13	0.00	0.15	0.71	0.00	0.220
W15R	HBA (α)	0.995	2.48	-7.16	17.7	13	0.00	0.12	0.75	0.00	0.218
E7V	HBB (β)	0.998	—	-7.17	17.8	13	0.00	0.11	0.75	0.00	0.217
H3L	HBB (β)	0.993	—	-7.26	17.4	13	0.01	0.10	0.74	0.00	0.211
D7G	HBA (α)	0.995	1.83	-7.25	17.6	13	0.00	0.08	0.75	0.00	0.204
A6P	HBA (α)	0.998	0.88	-7.25	16.2	13	0.00	0.08	0.69	0.00	0.189
V12I	HBB (β)	0.991	—	-7.22	17.1	13	0.01	0.01	0.72	0.00	0.175
<i>Wild-type reference:</i>											
WT	—	1.000	—	-7.21	17.9	13	—	—	—	—	—

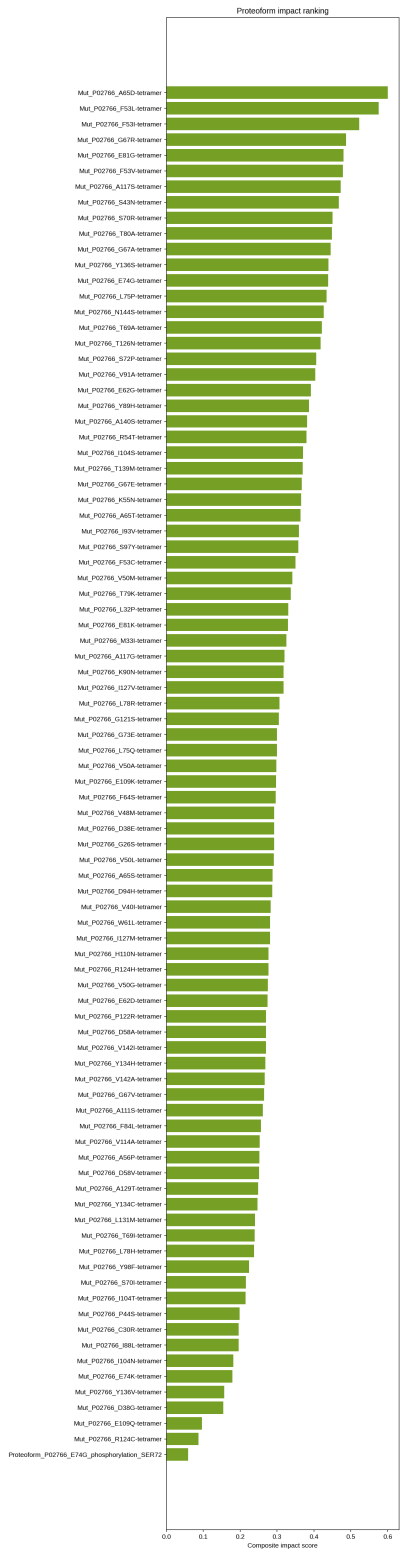


Figure 1 Complete Proteoform Impact Score ranking for the TTR missense-variant experiment (all scored proteoforms, sorted by composite score).

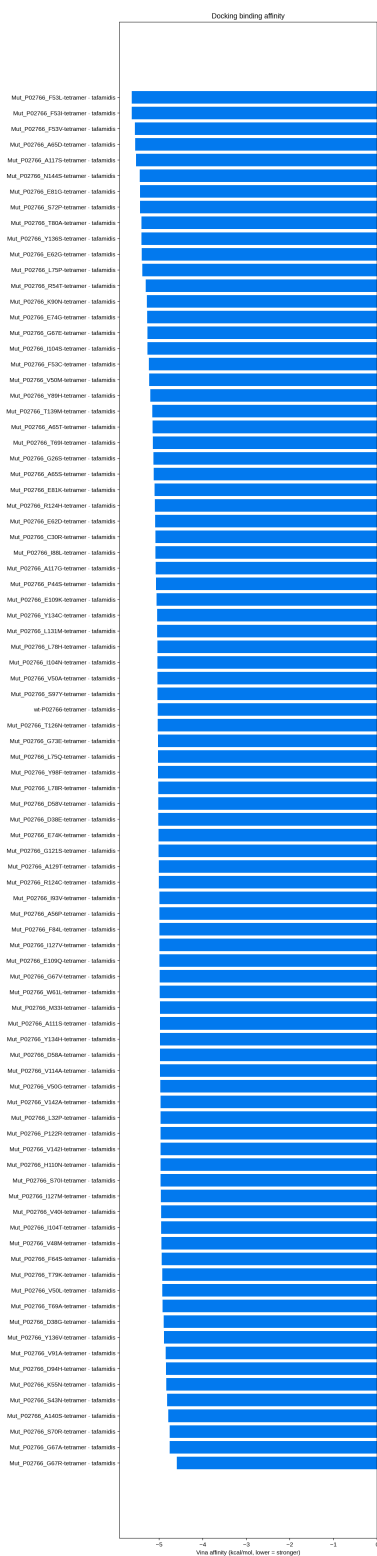


Figure 2 Complete tafamidis docking score distribution (AutoDock Vina) for the 88 docked TTR structures, including the wild-type tetramer.

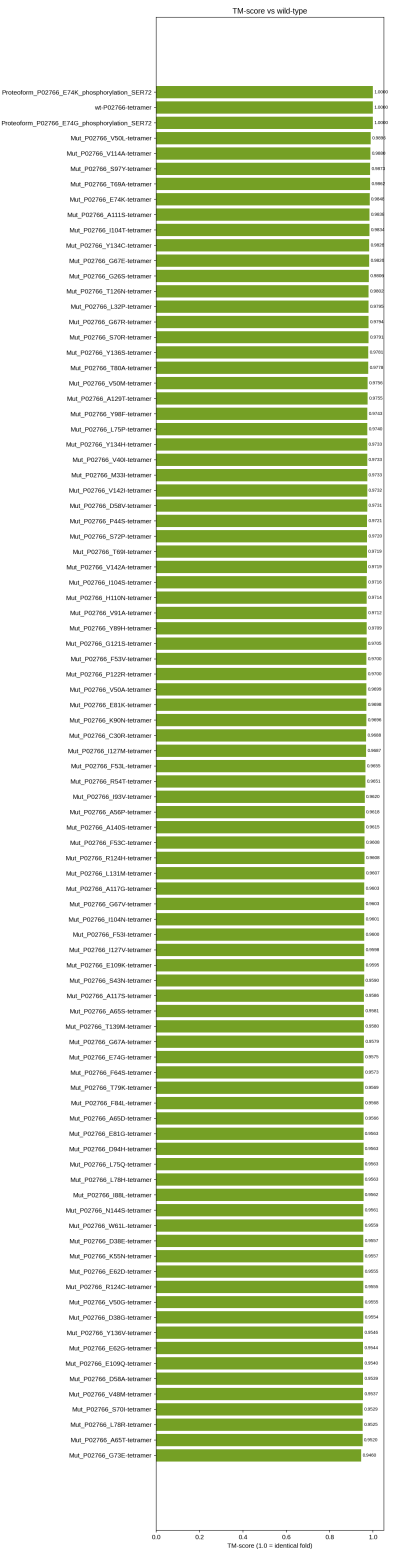


Figure 3 Complete TM-score summary for the 88 TTR structures against the wild-type reference.

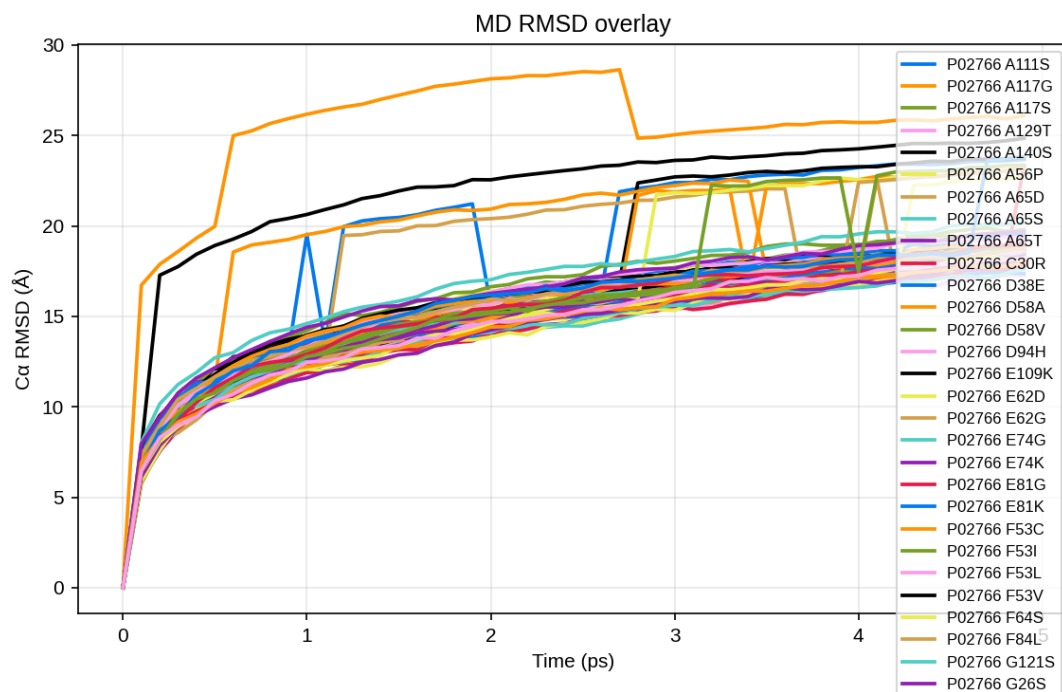


Figure 4 Backbone RMSD overlays of the short OpenMM validation trajectories for the TTR variant panel.

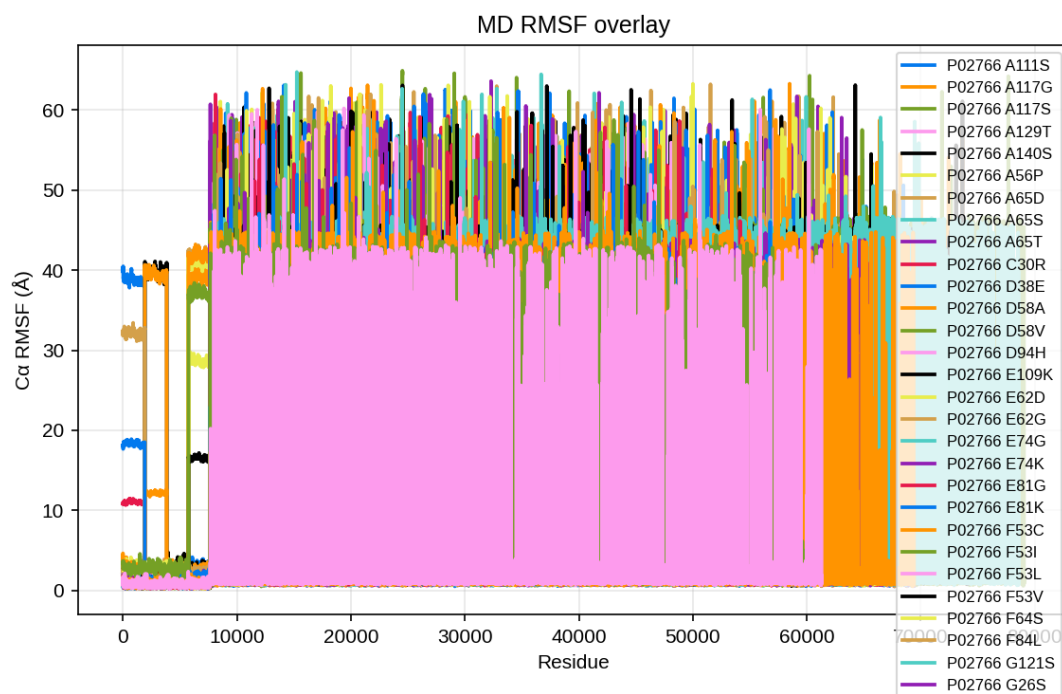
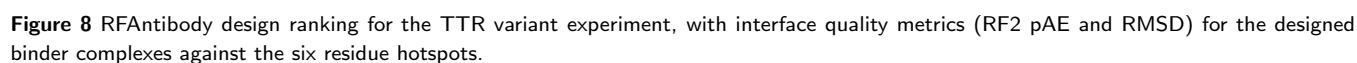


Figure 5 Per-residue RMSF overlays of the short OpenMM validation trajectories for the TTR variant panel.



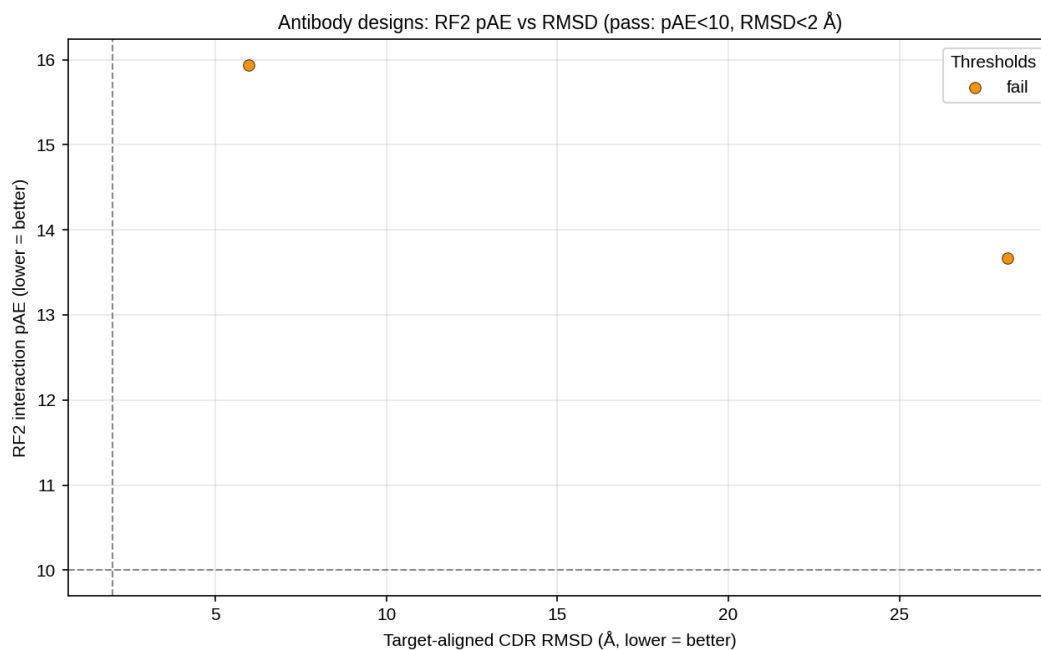


Figure 9 Interface PAE versus RMSD for the RFAntibody designs of the TTR variant experiment.



Figure 10 Louvain communities and betweenness centrality values of each residue in the V50M-TTR tetramer.

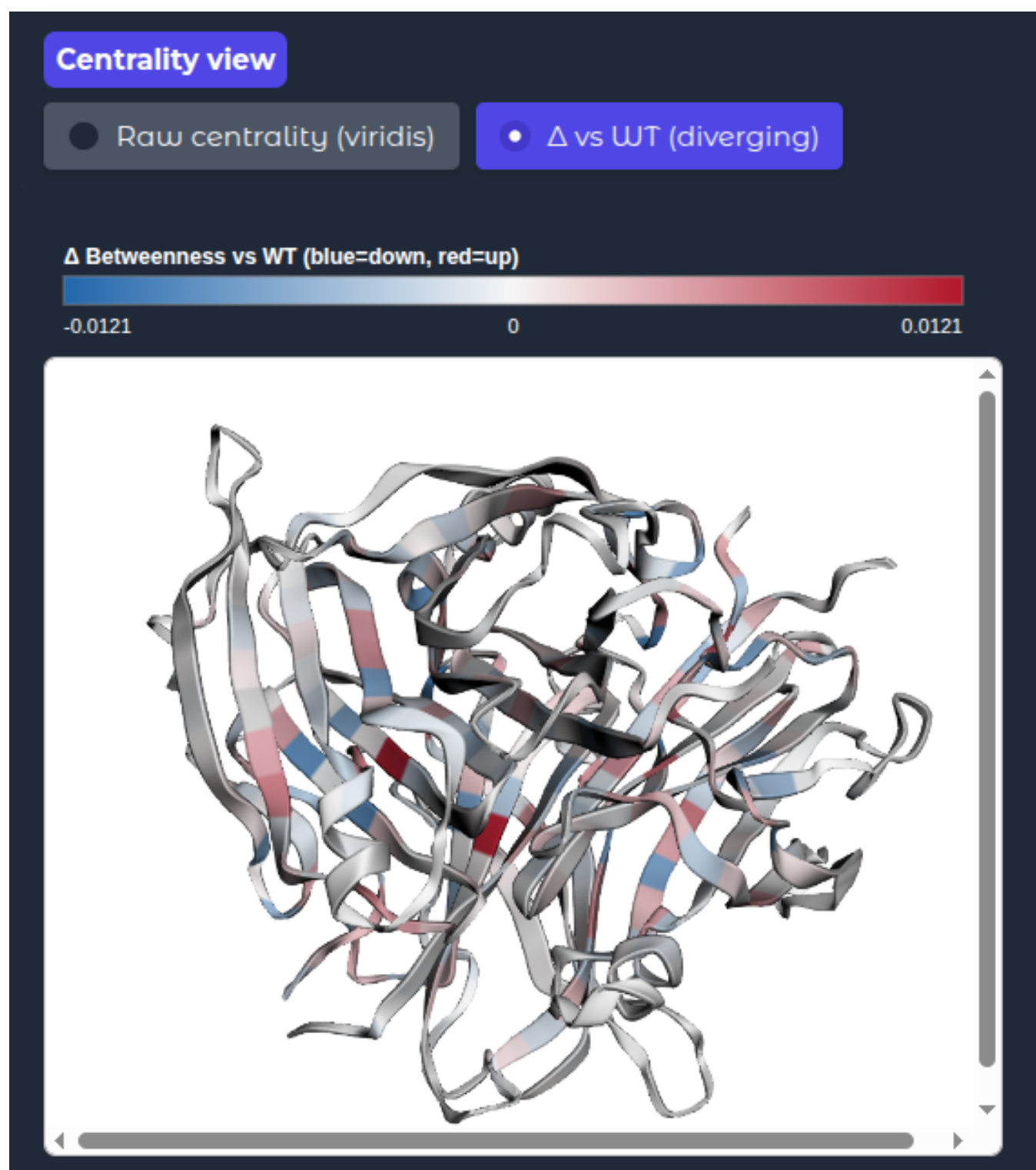


Figure 11 Residue-level differences in betweenness centrality values between V50M and wild-type TTR tetramers.

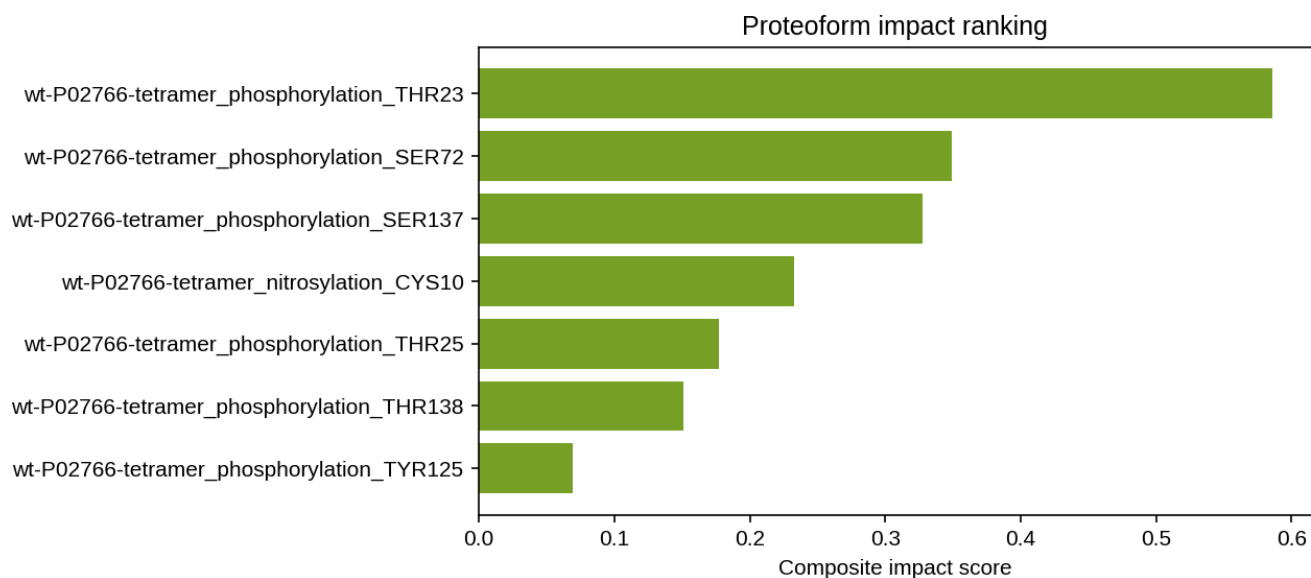


Figure 12 Proteoform Impact Score ranking for the TTR PTM experiment (all seven PTM proteoforms).

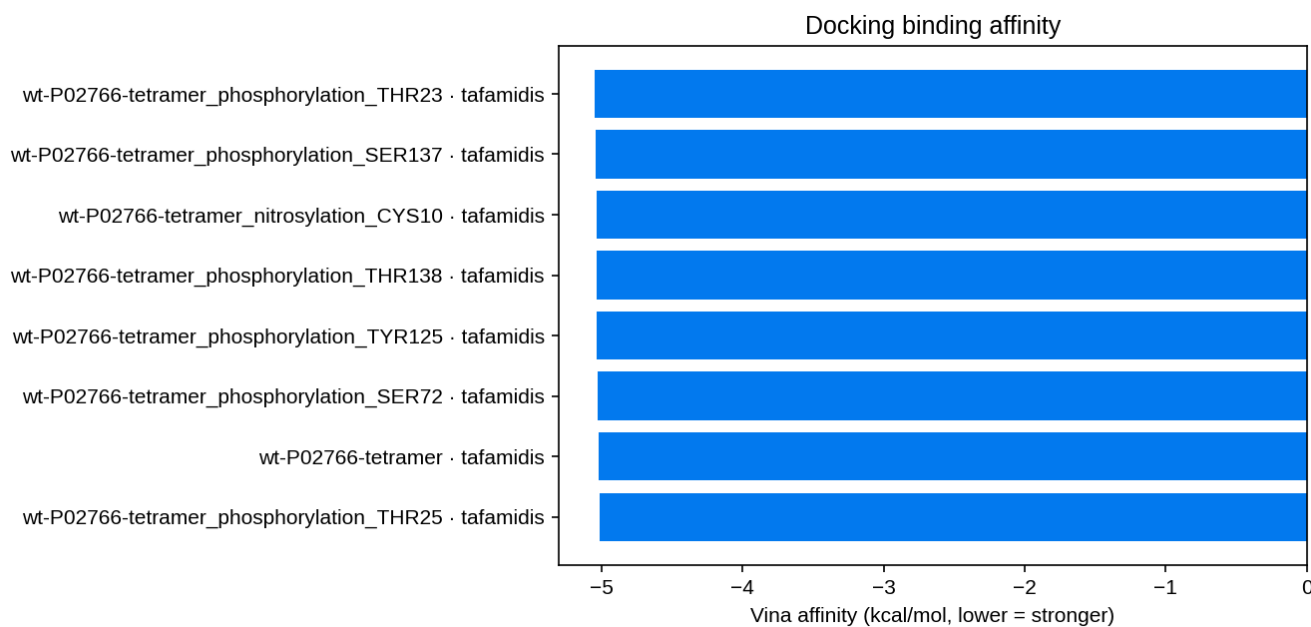


Figure 13 Tafamidis docking scores (AutoDock Vina) for the eight docked structures of the TTR PTM experiment, including the wild-type tetramer.

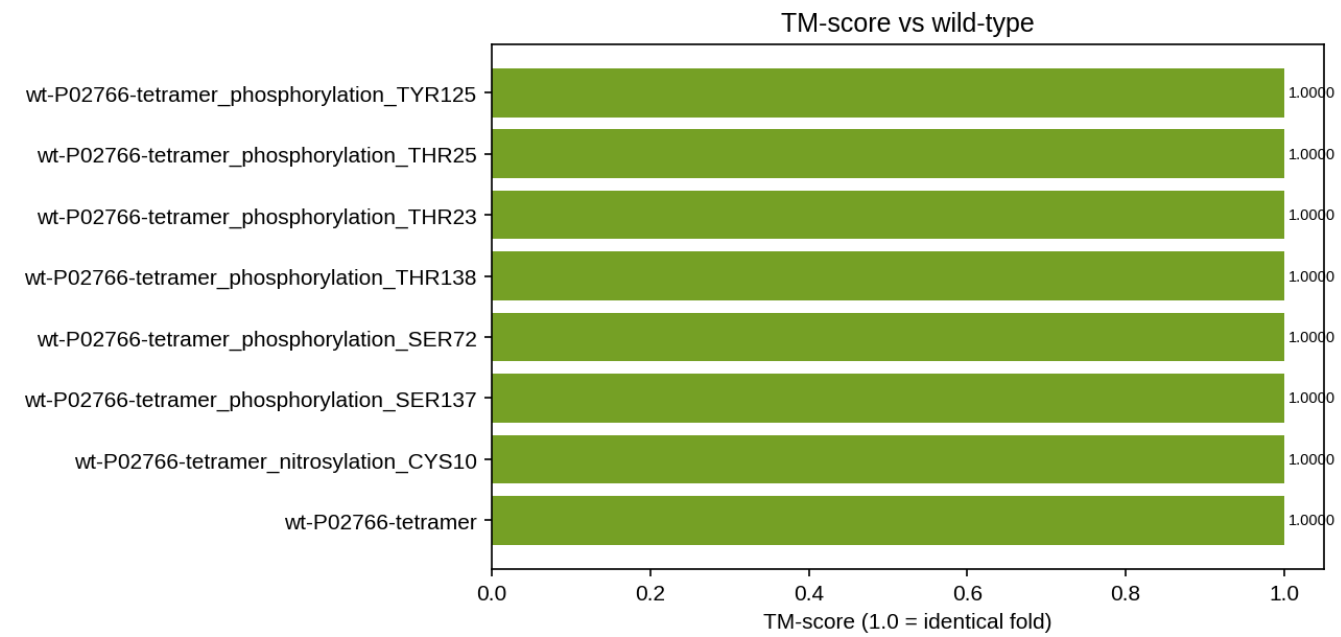


Figure 14 TM-score summary for the eight structures of the TTR PTM experiment against the wild-type reference.

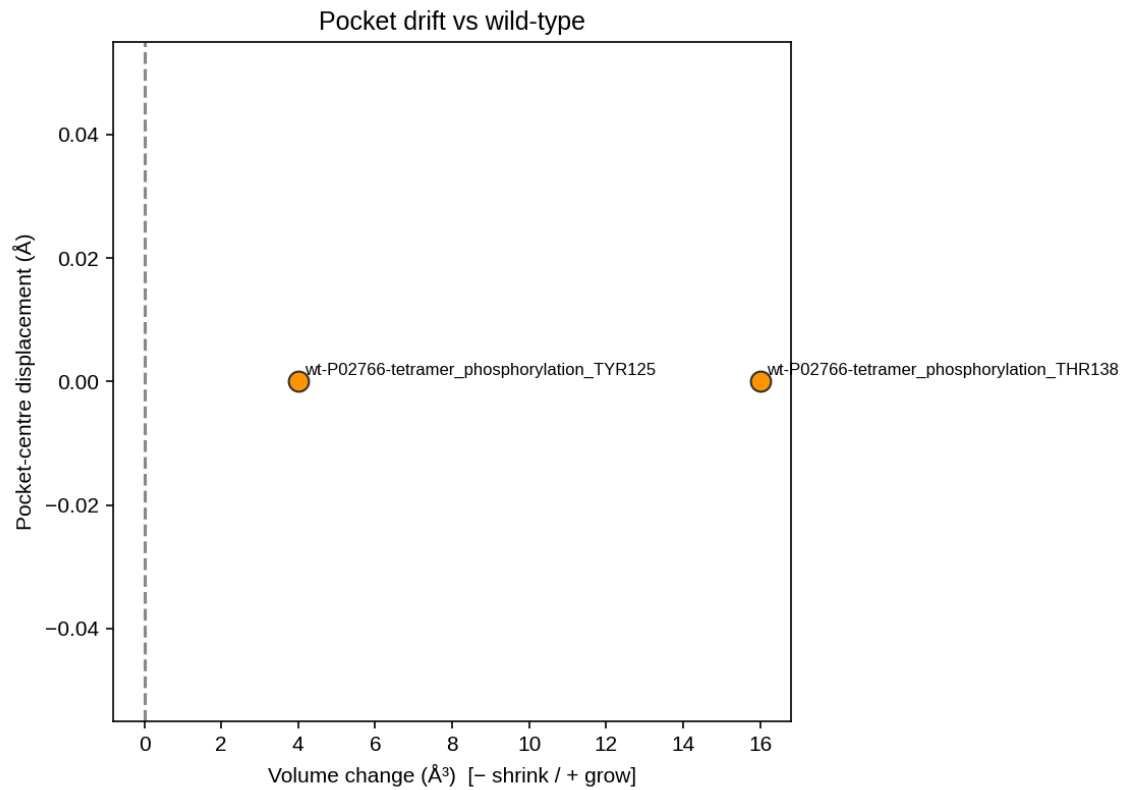


Figure 15 Binding-pocket drift for the TTR PTM proteoforms: pocket center displacement with respect to the wild-type reference.

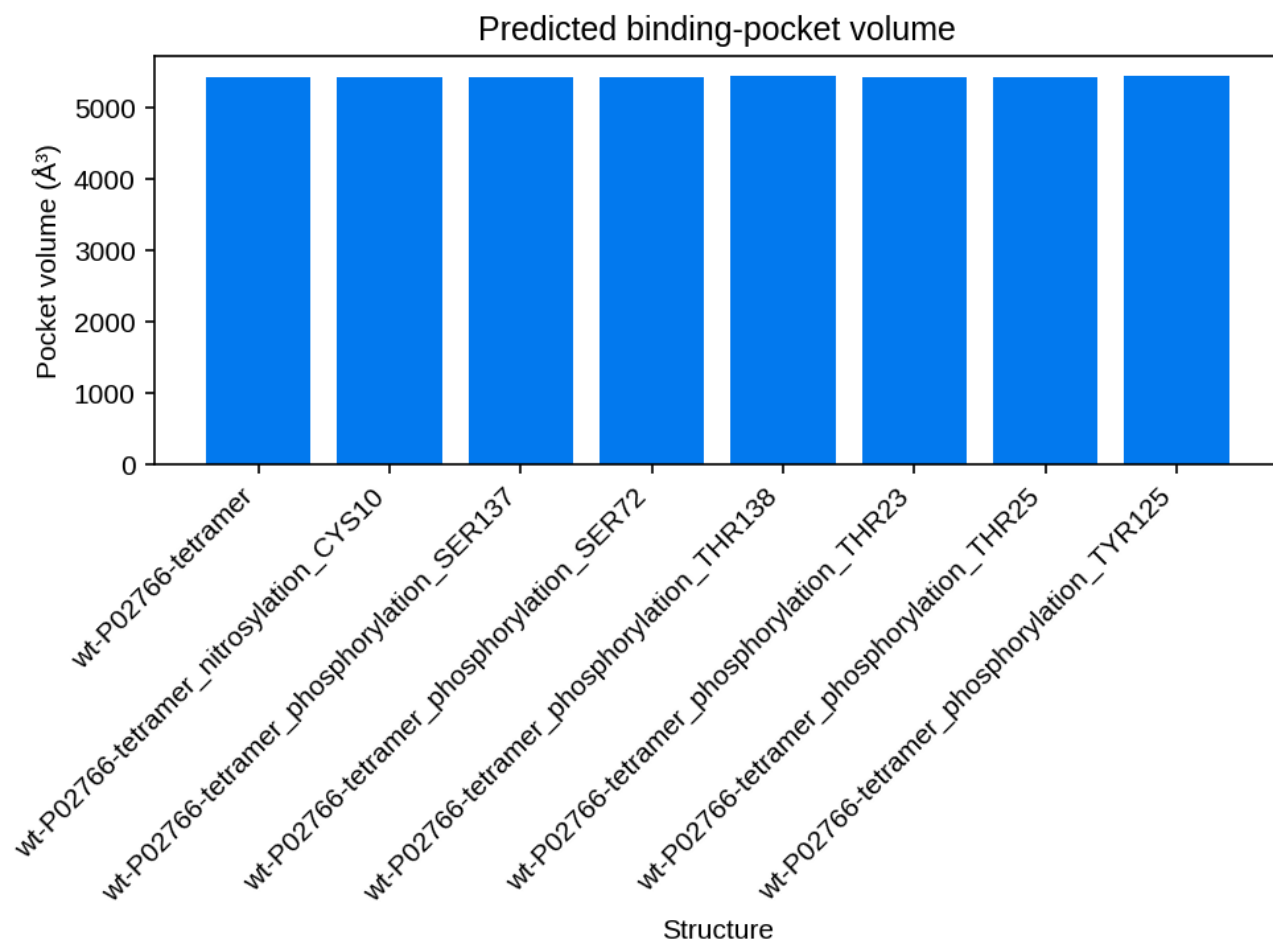


Figure 16 Binding-pocket volume for the TTR PTM proteoforms relative to the wild-type reference.

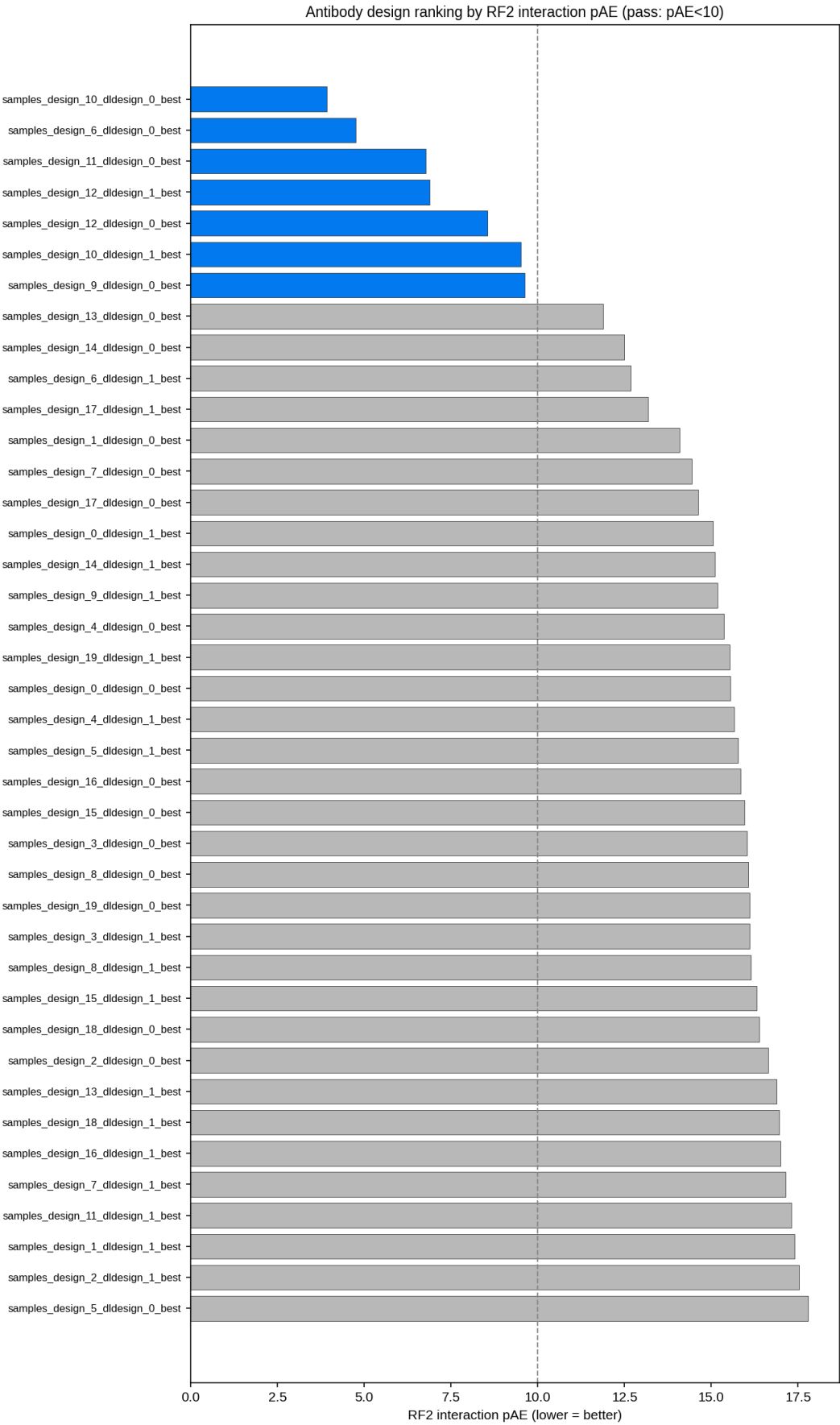


Figure 17 RFAntibody design ranking for the TTR PTM experiment, with interface quality metrics (RF2 pAE and RMSD) for the designed binder complexes against the six residue hotspots.

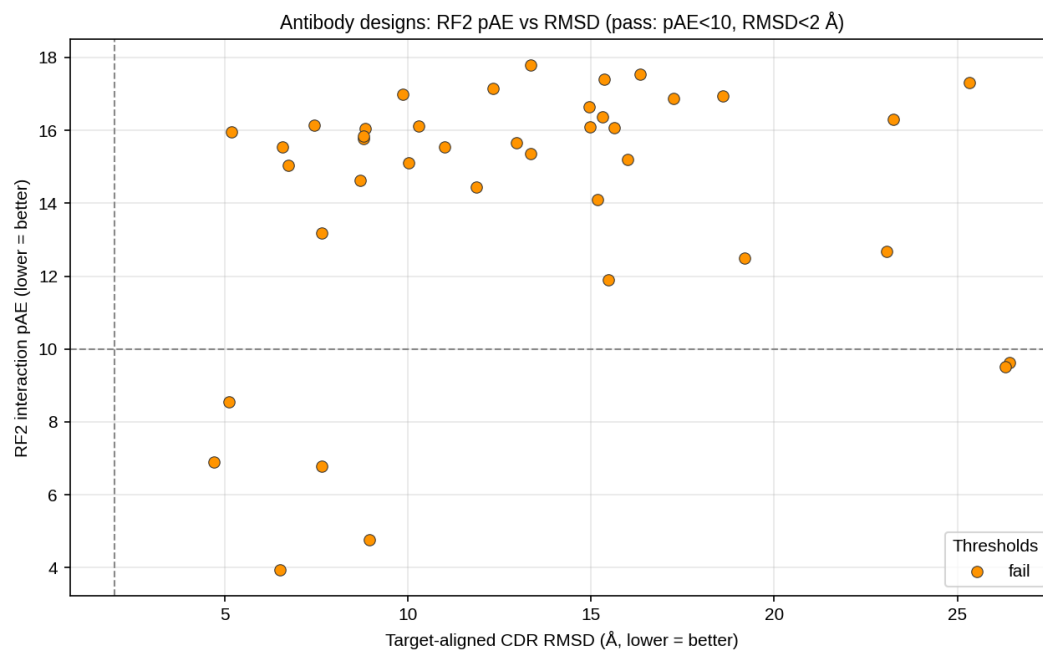


Figure 18 Interface PAE versus RMSD for the RFAntibody designs of the TTR PTM experiment.

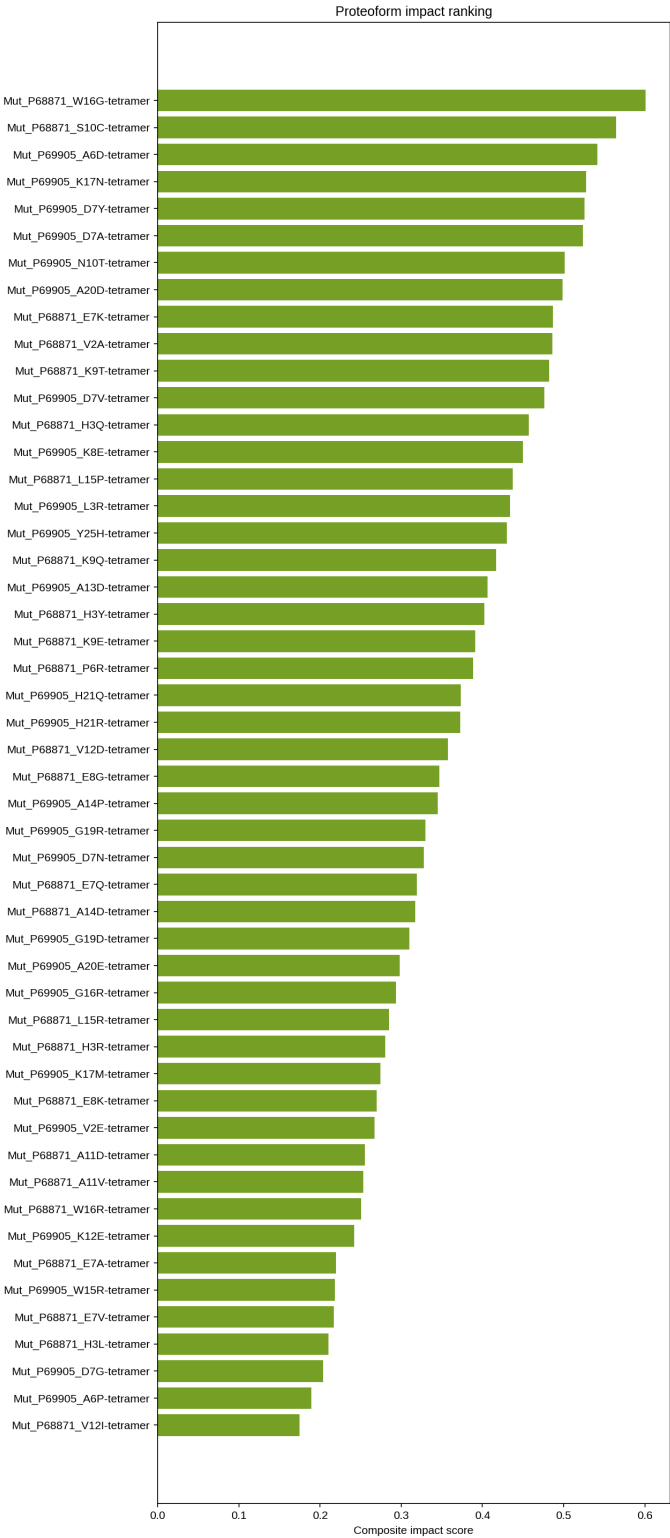


Figure 19 Complete Proteoform Impact Score ranking for the hemoglobin experiment (all scored proteoforms, sorted by composite score).

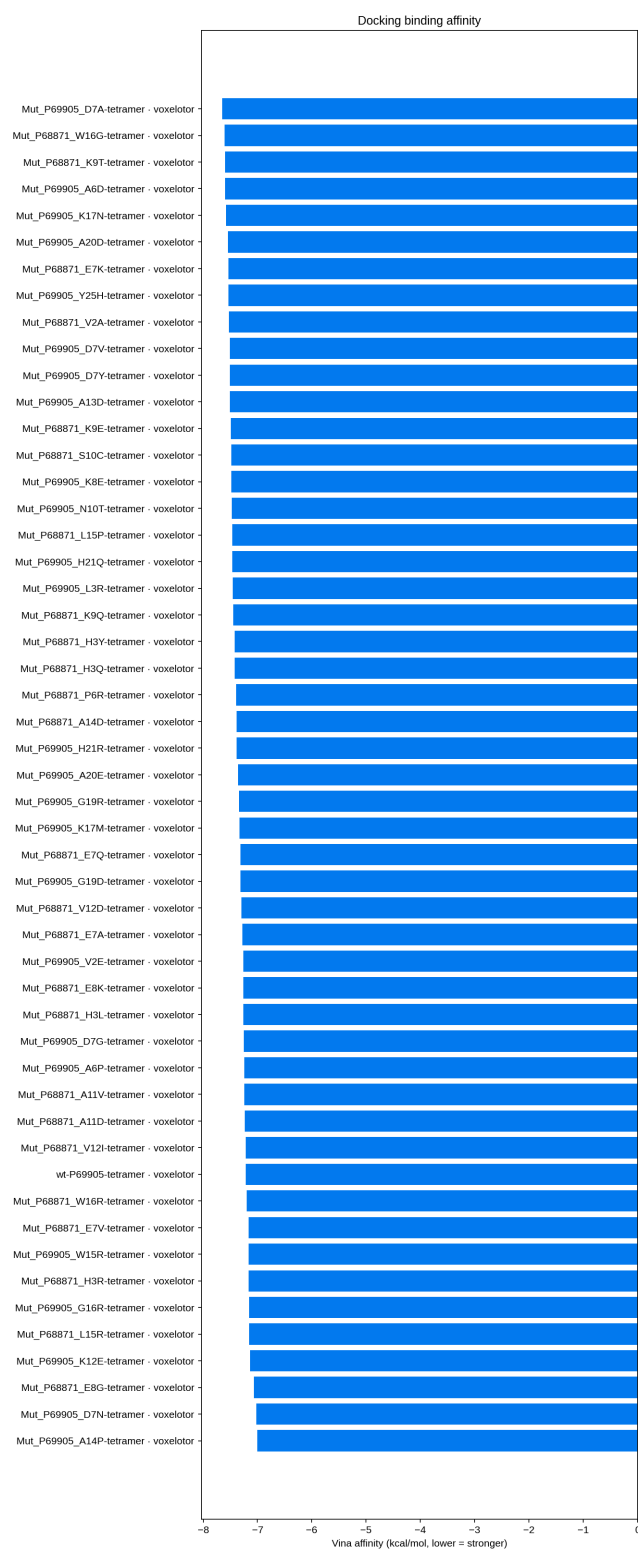


Figure 20 Complete voxelotor docking score distribution (AutoDock Vina) for the 51 docked hemoglobin structures, including the wild-type tetramer.



Figure 21 Complete TM-score summary for the 51 hemoglobin structures against the wild-type reference.

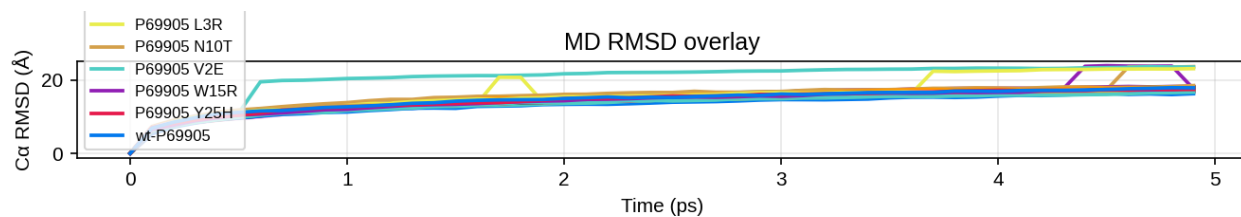


Figure 22 Backbone RMSD overlays of the short OpenMM validation trajectories for the hemoglobin variant panel.

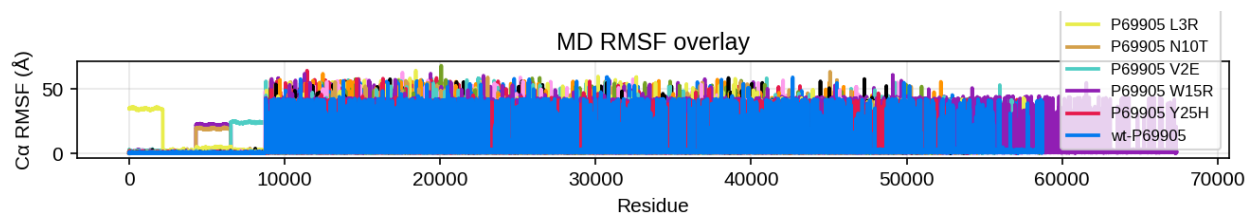
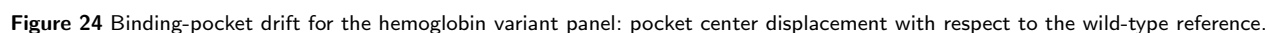


Figure 23 Per-residue RMSF overlays of the short OpenMM validation trajectories for the hemoglobin variant panel.



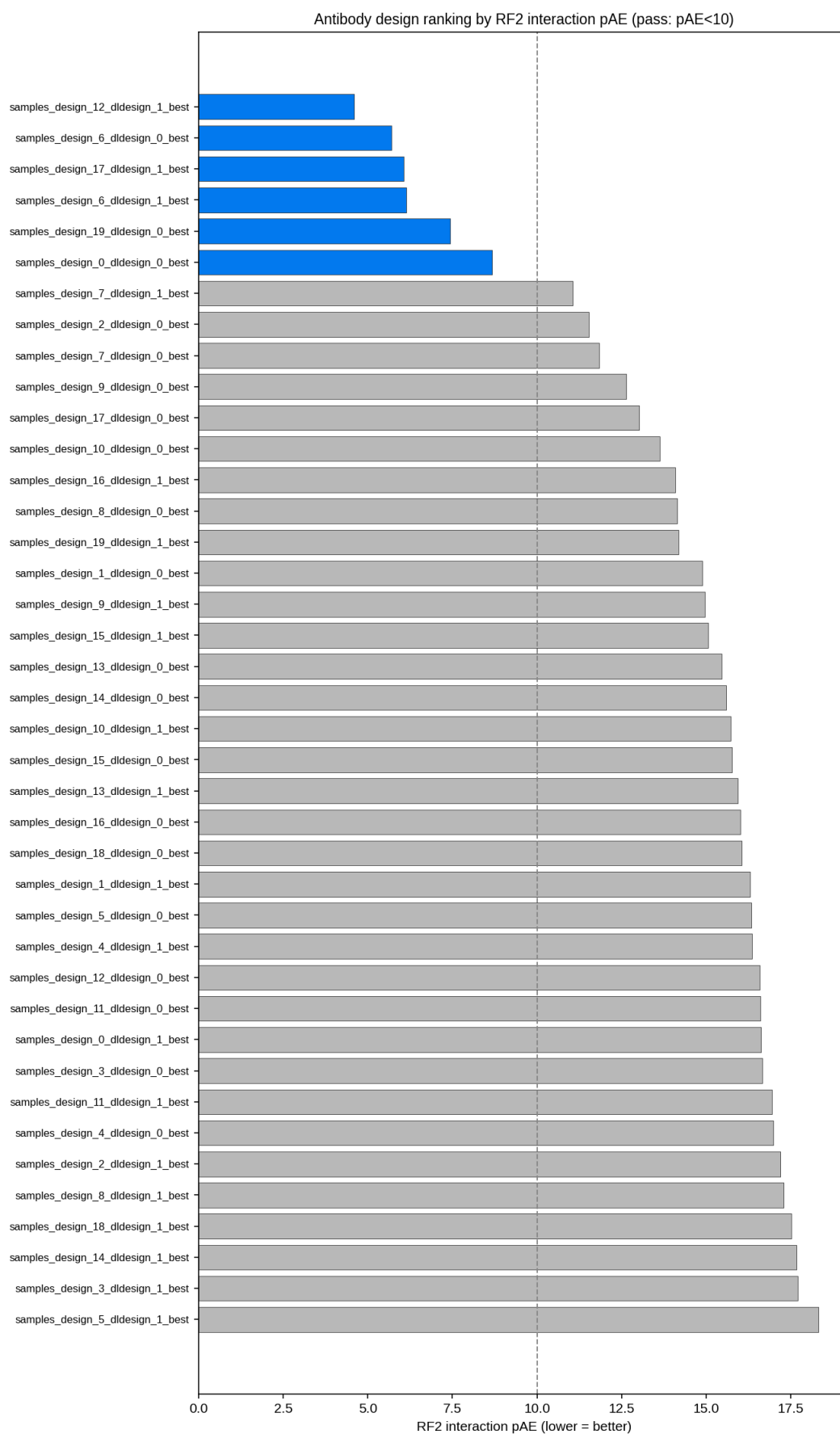


Figure 26 RFAntibody design ranking for the hemoglobin experiment, with interface quality metrics (RF2 pAE and RMSD) for the designed binder complexes.

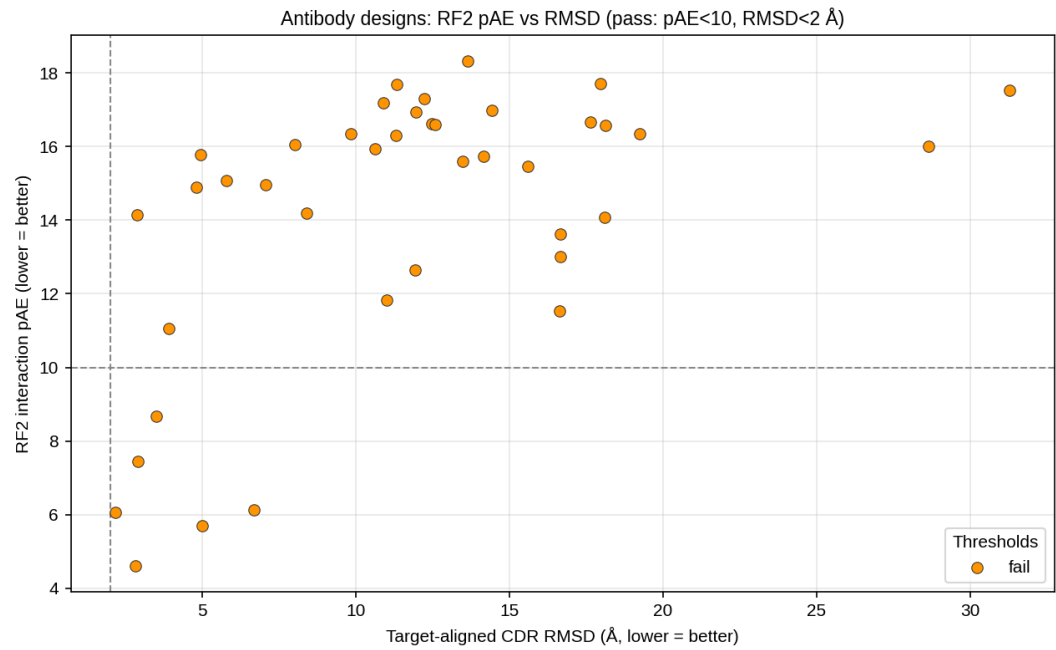


Figure 27 Interface PAE versus RMSD for the RFAntibody designs of the hemoglobin experiment.

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